Incorporating the Web into Machine Learning Framework for Natural Language Processing

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I. INTRODUCTION

Traditional written corpora for linguistics research are created primarily from printed text, such as newspaper articles and books. Such corpus-based approaches usually suffer from data sparse problem even in a very large corpus. With the growth of the World Wide Web as an information resource, there are many advantages to creating a corpus from the Web rather than printed text. All Web data is already in electronic form and therefore readable by computers, whereas not all printed data is available electronically. The vast amount of text available on the Web is a major advantage. Therefore, the Web is being increasingly used as an information resource in a wide range of NLP tasks [1], such as machine translation, question answering and language modeling. In our previous work in [2], we described how to use Web-based models for countability detection. Our approach shows the promising performance for the unsupervised system.

Most NLP tasks that have utilized the Web data have accessed it through search engines, using only the Web counts or examining a limited number of results pages. The tasks are reduced to determining n-gram probabilities which are then estimated by Web counts from search engine queries. This method only gathers information from the Web counts but does not require the computationally expensive downloading of actual text for analysis. Therefore, such simple Web-based methods have some limitations. Firstly, they rely on the assumption that page counts approximate n-gram frequencies. It certainly adds noise to the data. Another problem is that search engines were not designed for NLP research and the reported Web counts are subject to uncontrolled variations and approximations. Both Google[1] and Altavista[2] disregard punctuation and capitalization when matching a search term. This can lead to false positives, e.g., if the match crosses a sentence boundary. Also, the matches are likely to include links, web addresses, file names, and other non-textual data.

In this paper, we focus on eliminating the limitations of the Web counts and make an attempt to merge Web-based models and supervised approaches. We incorporate the Web counts as additional features into maximum entropy classifier to carry out the combination of the web with machine learning. The values of the features can be estimated by querying in the Web using some patterns. Such Web-based features are used to extract missing lexical and syntactic information in normal training corpus. Our approach is evaluated on two routine NLP tasks, which are zero anaphora resolution and article recovery.

II. MAXIMUM ENTROPY FRAMEWORK

Maximum Entropy (ME) is a general machine learning technique for estimating probability distributions from data. In ME framework, events are represented as multiplicities of weighted features. In training, the features' weights are derived based upon the distribution of the features in the training data, using an iterative algorithm. Weights are chosen to maximize the entropy of the model while minimizing the divergence between the model and the training data. In employing the model, events of a given context is evaluated by summing the weights of their representative features and normalizing over the context to obtain a probability distribution, as the following equation (1):

$$p(y \mid x) = \frac{1}{Z(x)} \exp\left[\sum_i \lambda_i f_i(x, y)\right] \qquad (1)$$

where $p(y|x)$ represents the probability of event y given context x, and λ_i represents the weight for feature f_i. $Z(x)$ is simply a normalizing factor to ensure a proper probability (equation (2)):

$$Z(x) = \sum_y \exp\left[\sum_i \lambda_i f_i(x, y)\right] \qquad (2)$$

We develop a classifier based on ME to determine whether a candidate is the correct resolution or not. For example, in zero anaphora resolution, given a candidate x ∈ {all possible candidates for a certain ZA}, we need to predict y ∈ {yes, no}, which classifies x as the correct antecedent or not. In article recovery, give a candidate x ∈ {a/an, the zero}, we need to predict y ∈ {yes, no}, which classifies x as the correct article or not. The primary work of modeling ME-based classifier is selection of a set of features (f_i) to be considered in the analysis of the training data and estimation of the appropriate weight (λ) to each feature. In the latter work, we use IIS, a hillclimbing algorithm to find the globally optimal weights given the training data. How to devise the regular features and Web-based features are determined according to the NLP tasks. We consider two tasks in this paper to investigate the combination of the Web and machine learning. The tasks are zero anaphora resolution and article recovery.

[1] http://www.google.com/
[2] http://www.altavista.com/

Table 1. The Web-based features for ZA resolution.

Type	Feature	Describe	Values
Web-based	SEM-CONSI	The rank of the semantic consistence of antecedent candidates.	1, 2, 3, 4…
	SEM-RELA	The semantic relations between the heads of the predicates in sentences containing ZA and an antecedent candidate.	transfer, happens-before, unknown

Table 2: The Web-based features for article recovery.

Types	Feature	Description	Values
Web-based	2-GRAM	Find the article that has the highest frequency co-occurring with the NP. (argmax(f(art, NP)))	a(an), the, zero
	3-GRAM	Find the article that has the highest frequency co-occurring with one word left to it and the NP. (argmax(f(w_1, art, NP)))	a(an), the, zero

III. ZERO ANAPHORA RESOLUTION

Zero anaphors (ZA) are generally noun phrases that are understood from the context and do not need to be specified. ZA resolution is important in many NLP tasks, such as machine translation, information extraction and question answering systems.

The regular features used in the ME-based classifier are motivated by previous researches on anaphora resolution. The feature set includes 4 grammatical, 5 lexical, 1 positional and 3 semantic features. The values of the features can be obtained directly from the corpus, the output of system modules and the semantic dictionaries with the help of heuristic rules.

The two Web-based features with their possible values are summarized in Table 1. Feature SEM-CONSI captures the semantic consistence computed from the Web. We consider one fact in ZA resolution, which is that the semantic constraint that is satisfied by a ZA, must be satisfied by its antecedent. That means that for a certain ZA, its antecedent and predicate must have semantic consistency. The values represent the rank of the consistence. For example, SEM-CONSI=1 means the current candidate has the highest semantic consistence among all the antecedent candidates. Feature SEM-RELA represents the semantic relations between the heads of the predicates of ZA and a candidate. The other fact in ZA resolution is that although the relation between a ZA and its antecedent is implicitly expressed, they are linked by a strong semantic relation between the heads of the predicates of the sentences containing the ZA and its antecedent. To describe such relations, three values are possible in our work. They are “transfer”, “happens-before” and “unknown”.

We trained and tested the ME-based classifier with the regular feature set and the combination of the regular set and the two additional Web-based features. The correct accuracies are 71.9% and 81.8%. We obtain 9.9 points improvements in comparison to the former classifier trained on the regular set.

III. ARTICAL RECOVERY

Article recovery is to decide when to use *a (an)*, *the*, or *zero* article at the beginning of a noun phrase (NP). Some languages, such as Chinese and Japanese, do not have any article or mark of the countability. Therefore recovering correct article in English translation is crucially important for high quality MT.

Our regular feature set consists of 10 common types of features. These features are motivated by previous researches on article recovery. They are the countability of heads, words of all words in NP, the two words left to NP, the words within the NP, the words following the NP, POS of head, POS of initial word in NP, POS of the word leftmost to NP, POS of all words in NP, POS of the word rightmost to NP. The values of the features can be obtained based on some heuristic rules, information in the corpus and the output of POS tagger.

The two Web-based features with their possible values are summarized in Table 2. For 2-GRAM, we estimate how frequently the NP led by an unknown article co-occurs with the, a/an, or zero. The value of the feature is the choice with the highest frequency in the web (f(art, NP)). For 3-GRAM, we take into account one word (w_1) to the left of the unknown article and the NP. The value of the feature is the article which has the highest co-occurring frequency in the web (f(w_1, art, NP)). To avoid data sparseness, in an initial query only the nominal or verbal heads are retained. Each NE is replaced by the corresponding common noun. (e.g., “SONY” -> “company”). Also, a set of inflected queries is generated by expanding a term into all its possible morphological forms. (e.g., “buy the book” -> “buy|bought|buying|buys the book”).

We trained and tested the ME-based classifier with the regular feature set and the combination of the regular set and the two additional Web-based features. The correct accuracies are 82.7% and 89.2%. We obtain 6.5 points improvements in comparison to the former classifier trained on the regular set.

IV. CONCLUSION

In this paper, we employ the web as additional knowledge resource to improve the performances of two NLP tasks. Unlike the previous research, we show how to combine the web and machine learning framework. In the ME framework, incorporating the web-based features into the regular features can improve the correct accuracy from 71.9% to 81.8% in ZA resolution and from 82.7% to 89.2% in article recovery.

REFERENCES

[1] Lapata, M. and Keller, F. 2005. Web-based Models for Natural Language Processing, ACM Transactions Speech and Language Processing (TSLP) 2(1), 1-31.

[2] Peng. J and Araki. K, “Detecting the Countability of English Compound Nouns Using Web-based Models”, In Companion Volume to the Proceedings of the Second International Joint Conference on Natural Language Processing (IJCNLP), pp.105-109, 2005.